ORTAM developer exercise

Candidate brief version 2

Build an employee-facing web application that helps an ORTAM employee prepare an asbestos assessment report. The employee enters building and inspection information. The application reads laboratory PDF reports with AI, matches each result to its sample location, and creates a report that the employee can review, preview and download in Word and PDF.

No prior asbestos experience is required. All technical interpretation and pricing rules needed for this exercise appear in the supplied documents. All buildings, samples, results and prices are fictional. This exercise does not qualify a real property for work or certify its safety.

What we want to assess

First, can you build a clear form and document workflow that an employee can use? Second, can you implement AI document analysis accurately, preserve its sources and let an employee correct mistakes? Floor plan mapping earns separate bonus points.

Reading order

1. Start with asbestos_exercise_walkthrough.pptx for the visual overview, or its PDF companion.

2. Read this brief. It defines the required behavior and submission.

3. Use the lab PDFs, employee field notes, pricing rules and Word master as your working inputs.

4. Consult implementation_guidance.pdf for optional public library references. Your app must run without access to ORTAM private code.

Time allowance

Plan for two working days, with a maximum of 16 focused hours for the core prototype. The optional mapping bonus can use up to four additional hours on a third day if ORTAM agrees before the exercise starts. Record actual time and identify unfinished work. ORTAM provides the deadline, AI access or budget, and submission channel separately.

AI coding assistants such as Codex, Cursor or Claude Code are allowed. Equivalent frameworks and libraries are acceptable. A locally running application is sufficient. The brief defines a prototype, not a production deployment.

The domain in plain language

Asbestos is a group of mineral fibres historically used in some building materials. In this exercise, the application records laboratory evidence about its presence. Candidates do not need to identify materials themselves or decide how real work should proceed.

An employee records where a material is located in a building and assigns a number to a sample of that material. A laboratory report lists that sample number and says whether the tested material contains asbestos. The software connects these two records using the sample number.

Term	Meaning in this exercise
Sample ID	A label such as S01. It links a laboratory result to the employee record. It is not a room number.
Material zone	One measured area or length of material, such as the ceiling in one office. It has one material ID, such as M03, and may have several samples.
Layer	One component within a sample. A floor sample can contain a tile layer and an adhesive layer with different results.
Detected or positive	The laboratory reports asbestos in that sample layer. Preserve the type and percentage if supplied.
Not detected or negative	The laboratory did not detect asbestos in the sampled layer. This does not describe every material in a building.
Inconclusive	The lab could not give a definite result. This remains unknown and requires follow up.
Unsampled or inaccessible	The employee has no lab result, for example because the area was locked. This remains unknown.
Removal and replacement estimate	A fictional budget scenario using the quantities and rates supplied for the exercise. It is not a contractor quotation.

The concentration percentage is a lab result, not an automatic risk ranking. Do not research or invent legal thresholds or remediation procedures. Follow the explicit exercise categories in fictional_rates_and_rules.pdf.

Example: the employee assigns S02 to the floor in corridor G01. The lab reports no asbestos in the tile but detects it in the adhesive. The app keeps both layer results and treats that floor material zone as detected.

Background reference: OSHA Asbestos Overview, https://www.osha.gov/asbestos . The supplied exercise rules govern this test, not an external regulatory standard.

What each input contains

File	How the candidate uses it
employee_field_notes.pdf	Enter this project and material information through the form you build. Automatic extraction of these notes is optional.
LAB-A_fictional_lab_report.pdf LAB-B_fictional_lab_report.pdf LAB-C_fictional_lab_report.pdf	Upload all three PDFs into the app. The AI extracts their sample and layer results. The PDFs have selectable text.
fictional_rates_and_rules.pdf	Use these fixed rates, review categories, calculations and exclusions. Implement arithmetic in code.
master_asbestos_report.docx	Use its report structure for editable Word output. Its bracketed text describes fields. You may adapt the placeholders to your chosen generation library.
floorplan_basement.pdf floorplan_ground.pdf	Optional bonus inputs. Employees upload a PDF plan and manually place sample markers.
optional_sample_positions.pdf	Suggested locations for demonstrating the mapping bonus. These positions are illustrative.
implementation_guidance.pdf	Optional public tools and architecture references. These are suggestions, not dependencies you must use.

Who supplies which information

The employee supplies the building details, exact room or location, sample IDs, material descriptions, quantities, units, condition and planned disturbance. Planned disturbance means the employee expects work to affect that material, such as replacing a floor.

The lab PDFs supply the sample ID, layer description, detected or not-detected result, asbestos type, percentage and any inconclusive-result note. A lab result does not supply the site quantity or establish the physical condition of an entire material zone.

The app combines the accepted employee and lab information. It drafts findings and review recommendations using the supplied rules, calculates the exercise estimate, and creates the two report formats.

Required employee workflow

1 Project and material form

Let an employee create a project, enter the client and building, inspection and report dates, inspection scope, planned work and limitations. Let them add, edit and remove material zones. Each zone needs a material ID, description, floor, precise location, one or more sample IDs, quantity, unit, condition, planned disturbance and the supplied rate reference when available.

Use the supplied condition values intact, damaged, poor or unknown. Planned disturbance is a yes or no field. A rate reference such as R_PIPE links the zone to a row in the supplied rate card. Show a rate or unit mismatch for review instead of applying an unrelated price.

Support square metres and linear metres exactly as supplied. Allow unknown quantities without replacing them with zero. Save the project so it survives a page reload and can be reopened.

2 Laboratory upload and AI extraction

Accept the three PDF lab reports. Extract their text or pages and send the content to an AI model. Return structured sample and layer records. Show extracted values beside a source reference including file, page and supporting text. An employee can correct the extracted fields before accepting them.

Display processing and failure states. A failed AI request must not look like a completed analysis. Reimporting the same PDF must not duplicate findings or costs.

3 Matching and employee review

Match records within the project using sample IDs. Preserve every layer result. Show unmatched IDs, contradictory records, missing quantities, inconclusive results and unsampled areas. Require an employee to resolve a conflict or leave it explicitly flagged in the draft. Do not silently guess an ID match.

4 Editable draft report

Create the report sections in the supplied Word master. Explain what was found, where it was found, the source sample and lab report, suggested professional review actions, costs and unresolved limitations. Let the employee edit report narrative and correct project information before regenerating the output.

5 Preview and downloads

Generate an editable Word DOCX and a PDF from the same accepted report revision. Display that PDF in the app as the report preview. Provide working Word and PDF download actions. After an edit, label the preview outdated until regeneration completes. Both downloads must show the same revised facts and cost totals.

A full Word editor in the browser is unnecessary. Employees can edit form fields and narrative in the app, then continue editing the downloaded Word file in Microsoft Word.

Interpretation and report content

The AI responsibilities

Use AI for extracting the laboratory PDF content and drafting explanations based on accepted evidence. AI-generated narrative must identify locations and source samples and follow the exercise rules. Show it as a draft for employee review. The candidate must demonstrate an actual AI call, not a hardcoded table of supplied results.

The application responsibilities

Validate the extracted record structure, match sample IDs, preserve layer results and calculate costs in code. Keep quantities on material zones. Two samples from one zone do not create two areas to charge. Missing facts stay missing until an employee supplies them.

Exercise category	Rule and expected wording
A Priority professional review	Detected material with poor or damaged condition, or with planned disturbance. Explain why a professional should review it before affected work.
B Professional management review	Other detected material. Flag it for professional management review without automatically prescribing removal.
C Resolve missing evidence	Inconclusive, inaccessible or unsampled material. Say what information is missing and retain the uncertainty.
D Record sampled finding	Not detected in the sampled material. Record its limited scope and omit an asbestos-specific cost for it in this exercise.

Required report sections

Project information and scope. Executive summary. A record for every material zone with location, condition, sample IDs, layer findings and lab sources. Draft recommendations and their reasons. Missing information and inspection limitations. The illustrative cost scenario with assumptions and exclusions. Employee review status and any plan appendix.

These categories are invented for this software exercise. The report must not claim legal compliance, building-wide absence of asbestos or clearance. No airborne exposure or clearance test data is supplied.

Costs and expected behavior

The fictional costing scenario

Price full removal and replacement only for confirmed detected zones with known quantities and compatible rates. This scenario does not itself recommend that every detected material should be removed. Keep confirmed but unmeasured material visibly marked Not priced.

Zone amount = quantity x (removal rate + replacement rate). Add all priced zone amounts. Add mobilization of CAD 2500 and a professional assessment allowance of CAD 1200 once per project when there is at least one priced zone. Add contingency equal to 20 percent of that direct subtotal. Show amounts in CAD to two decimals before tax.

Separate removal and replacement line amounts, fixed allowances, contingency and total. List exclusions from the supplied rules, including unmeasured and unresolved material. Validate quantity and unit changes, then recalculate the estimate. Do not invent a quantity or price.

Simple arithmetic example

For an unrelated example zone of 10 square metres with removal at CAD 50 per square metre and replacement at CAD 20 per square metre, the zone amount is CAD 700. With the two project allowances, the direct subtotal is CAD 4400. Contingency is CAD 880 and the illustrative total is CAD 5280 before tax. This example teaches the formula and is not the answer to the supplied building case.

Case details to handle correctly

Case	Expected app behavior
S02 floor sample	Keep the not-detected tile result and detected adhesive result. The zone is detected.
S03 and S07 ceiling samples	Keep both samples but charge the single M03 zone quantity of 140 square metres once.
S05 exterior panel	Detected with a missing quantity. Show it in the report and mark its cost Not priced.
S06 and M07	S06 is inconclusive. M07 was inaccessible and unsampled. Both remain unresolved.
S04 plaster sample	Record not detected only for the sampled plaster. Do not generalize to the whole building.

Acceptance and submission

Core completion checklist

The employee can enter all supplied field notes in a clear form, edit a material zone, save and reload the project.

The app ingests all three lab PDFs and uses a real AI model. It preserves sample IDs, individual layers, result labels, percentages where supplied, and file and page evidence.

The employee can review and correct extracted fields. Unknown IDs and contradictory results appear visibly for review. Missing quantities remain unknown.

The report follows the master structure, explains affected locations, gives grounded draft review actions and includes every unresolved limitation.

Costs follow the supplied rules. Multiple samples in one zone do not duplicate charges. Changing a quantity updates the amounts.

The preview works. Both the editable Word and PDF downloads open and contain the same accepted report revision. Edits cannot silently leave the preview current when it is stale.

Uploading the same lab PDF again does not create duplicate findings or costs. AI or export failures display an understandable error and allow retry.

What to submit

Provide source code as a repository link or ZIP, setup instructions with prerequisites, a short demo recording or screenshots showing the workflow and preview, one generated Word report and its matching PDF, and a short handover note. In the note, state actual time spent, AI tools used, what you checked yourself, incomplete features and known limitations.

Keep model credentials on the server and out of source control and reports. Include an environment variable example without real secrets. Do not require access to private ORTAM repositories. No production deployment is required.

Scoring

Core score out of 100: employee workflow 25, AI lab analysis 30, report quality and recommendations 20, calculations 15, and engineering and handover 10. The mapping bonus adds up to 10 separately recorded points and cannot compensate for incorrect core findings.

Optional mapping and scope limits

The floor plan bonus

Let an employee upload a PDF floor plan, place and move a marker manually, and assign a sample ID. Display the matched laboratory status beside that marker using a label or symbol as well as color. The S02 marker must reflect the detected adhesive even though its tile layer is negative. Inconclusive and unmatched samples need distinct unresolved statuses.

Store the PDF file, page, marker position and sample ID. Keep marker positions aligned when zooming or resizing and preserve them when reloading the project. Use the supplied sample positions for the demo. S05 is on an exterior elevation and does not appear on either interior plan. Include a labeled plan or marker register in the report if time allows.

The supplied floor plans are fictional and not to scale. Do not calculate real dimensions from them. PDF-to-CAD conversion, CADCraft integration, automatic room recognition and automatic marker placement are separate research ideas. They are not requirements for this bonus.

Suggested two day plan

Day 1: review the inputs, build the project and material form, save a project, upload the lab PDFs, connect AI extraction and show employee review of the extracted records.

Day 2: complete matching and rules, generate editable report content, implement deterministic costs, create Word and PDF export, show the preview and check the supplied cases. Leave time for setup instructions and the demo.

Optional third day: up to four additional hours for PDF plan mapping if agreed before the test. Finish the core before adding the bonus. Stop at the agreed limit and describe what remains.

Outside the core exercise

No asbestos research or specialist qualification. No legal interpretation. No scanned-PDF OCR. No full browser-based Word editor. No CAD conversion. No enterprise authentication, multi-user permissions, billing system, real customer data, production hosting or branded pixel-perfect interface.

A working, understandable prototype with correct results is the target. Candidates may explain reasonable implementation choices in the handover. The requirements above take precedence if a suggested tool or implementation note conflicts with them.